

Appendix A: Derivation of Equivalence Between Estimators

Gregory E. Paradis

Appendix A: Derivation of Equivalence Between Estimators

This appendix presents a rigorous derivation showing that the weighted estimator introduced in the main manuscript is equivalent to the size-biased estimator of Van Deusen [1986], Ducey and Gove [2015]. Throughout, we assume a fixed binning scheme for DBH classes $\mathcal{I}$ with midpoints x_i and employ the notation defined here.

Preliminaries

Let X denote diameter at breast height (DBH) with underlying PDF $f(x; \boldsymbol{\theta})$ and CDF $F(x; \boldsymbol{\theta})$ parameterised by $\boldsymbol{\theta} \in \Theta$. Under horizontal point sampling (HPS) with basal area factor C_{BA} , the inclusion probability of a stem of DBH x is proportional to its basal area, i.e., $\pi(x) \propto x^2$. The size-biased PDF of order two is therefore

$$(A1) \quad f_{\text{sb}}(x; \boldsymbol{\theta}) = \frac{x^2 f(x; \boldsymbol{\theta})}{\kappa(\boldsymbol{\theta})}, \quad \kappa(\boldsymbol{\theta}) = \int_0^\infty x^2 f(x; \boldsymbol{\theta}) \, dx,$$

which is well-defined whenever $\kappa(\boldsymbol{\theta}) < \infty$.

Let t_i denote the observed HPS tally (aggregated across plots) for bin i , and let $\hat{y}_i$ denote

14 the expanded stand table value

$$(A2) \quad \hat{y}_i = C_{BA} \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} t_{ij} \bar{g}_{ij}^{-1}.$$

15 Define the HPS expansion factor

$$(A3) \quad f_E(x; C_{BA}) = \frac{40000 C_{BA}}{\pi x^2},$$

16 and the compression factor $f_C(x; C_{BA}) = f_E(x; C_{BA})^{-1}$.

17 **Reference estimator**

18 The size-biased estimator minimises the sum of squares between observed tallies and the
19 size-biased PDF evaluated at bin midpoints:

$$(A4) \quad Z_C(\boldsymbol{\theta}, s) = \sum_{i \in \mathcal{I}} \left[t_i - s f_{\text{sb}}(x_i; \boldsymbol{\theta}) \right]^2,$$

20 where s is a scaling parameter converting continuous densities to discrete bin counts. The
21 minimiser $(\hat{\boldsymbol{\theta}}_C, \hat{s}_C)$ is obtained by solving the normal equations associated with (A4) using
22 standard non-linear least squares algorithms.

23 **Weighted estimator**

24 The weighted estimator operates on expanded stand table data with weights matching the
25 inverse of the expansion factor:

$$(A5) \quad Z_T(\boldsymbol{\theta}, s) = \sum_{i \in \mathcal{I}} w_i^2 \left[\hat{y}_i - s f(x_i; \boldsymbol{\theta}) \right]^2, \quad w_i = f_C(x_i; C_{BA}).$$

26 Recall $\hat{y}_i = f_E(x_i; C_{BA}) t_i$ by construction.

Key lemma

Lemma 1. *For all $\boldsymbol{\theta} \in \Theta$ and $s > 0$, the objectives $Z_C(\boldsymbol{\theta}, s)$ and $Z_T(\boldsymbol{\theta}, s')$ differ by a positive multiplicative constant after an appropriate reparameterisation of s .*

Derivation. Insert $\hat{y}_i = f_E(x_i; C_{BA})t_i$ and $w_i = f_C(x_i; C_{BA})$ into (A5):

$$(A6) \quad Z_T(\boldsymbol{\theta}, s) = \sum_{i \in \mathcal{I}} [f_C(x_i; C_{BA})f_E(x_i; C_{BA})t_i - f_C(x_i; C_{BA})sf(x_i; \boldsymbol{\theta})]^2$$

$$(A7) \quad = \sum_{i \in \mathcal{I}} [t_i - sf_C(x_i; C_{BA})f(x_i; \boldsymbol{\theta})]^2.$$

Using the explicit form of f_C , we have

$$(A8) \quad f_C(x; C_{BA})f(x; \boldsymbol{\theta}) = \frac{\pi}{40000C_{BA}}x^2f(x; \boldsymbol{\theta}) = \frac{\kappa(\boldsymbol{\theta})}{40000C_{BA}}f_{sb}(x; \boldsymbol{\theta}).$$

Let $s' = s\kappa(\boldsymbol{\theta})/(40000C_{BA})$. Then

$$(A9) \quad Z_T(\boldsymbol{\theta}, s) = \sum_{i \in \mathcal{I}} [t_i - s'f_{sb}(x_i; \boldsymbol{\theta})]^2 = Z_C(\boldsymbol{\theta}, s').$$

Hence Z_T and Z_C share the same level sets up to a bijective scaling of s .

Equivalence of minimisers

Let $(\hat{\boldsymbol{\theta}}_C, \hat{s}_C)$ minimise (A4). By the lemma, there exists a corresponding scaling $\hat{s}_T = \hat{s}_C 40000C_{BA}/\kappa(\hat{\boldsymbol{\theta}}_C)$ such that $(\hat{\boldsymbol{\theta}}_C, \hat{s}_T)$ minimises (A5). Conversely, any minimiser of (A5) maps back to a minimiser of (A4) by the inverse scaling. Therefore the estimators produce the same parameter estimates $\hat{\boldsymbol{\theta}}_C = \hat{\boldsymbol{\theta}}_T$ and the fitted PDFs align after transforming between tally and stand table space.

Extensions

The argument extends directly to unbinned data by replacing discrete sums with integrals. Moreover, any objective function proportional to the squared error (e.g., weighted least squares with constant variance within space) inherits the same equivalence. Likelihood-based estimators such as Poisson regression also admit analogous derivations: incorporating Horvitz–Thompson weights in the log-likelihood yields the same score equations as using size-biased PDFs.

Implications

Because the minimisers coincide, diagnostic quantities (RSS, chi-square statistics, fitted curves) are the same up to deterministic re-scaling between tally and stand table space. In practice, the weighted formulation provides a computationally convenient alternative that makes use of standard PDF implementations and alleviates the need for bespoke size-biased forms.

References

- Mark J. Ducey and Jeffrey H. Gove. Size-biased distributions in the generalized beta distribution family, with applications to forestry. *Forestry*, 88(2):143–151, 2015. doi: 10.1093/forestry/cpu038.
- Paul C. Van Deusen. Fitting assumed distributions to horizontal point sample diameters. *Forest Science*, 32(1):146–148, 1986. doi: 10.1093/forestscience/32.1.146.